DATE Functions

Question 1

Add 6 months to each employees hire date using DATEADD()

Syntax

```
Select emp_id,
        name,
        hire_date,
DATEADD(month, 6, hire_date) AS hire_plus_6_months
From Employees;
```

emp_id	Name	hire_date	hire_plus_6_months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Question 2

Use DATEDIFF() to find age in days from dob to today

~~using today~~ Using today = 2026-04-22

Syntax

```
Select Student_id,
        Name,
DATEDIFF(day, dob, current_date) AS age_in_days
From Students;
```

Student_id	Name	age_in_days
101	Maya	7569
102	Ethan	7816
103	Sienna	7349

Question 3

Find how many days are left until each event using DATEDIFF()

```
Select Event_id,
       Event_Name,
  DATEDIFF(day, current_date, event_date) AS days_remaining
FROM Events;
```

Event_id	Event_name	days_remaining
1	Seminar	-312
2	Workshop	-233
3	Hackathon	-458

(Negative = Event already passed)

Question 4

Calculate the number of days between issue_date and due date

Syntax

```
Select invoice_id,
       issue_date,
       due_date,
  DATEDIFF (day, issue_date, due_date) AS days_between
  from invoices;
```

Invoice_id	issue_date	due_date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Question 5

Format start_date as 'Month YYYY' using To_Char()

Syntax

```
Select course_id,
        name,
    TO_Char(start_date, 'Month YYYY') AS formatted_date
  from courses;
```

Course_id	name	formatted_date
201	SQL Basics	May 2025
202	Python	June 2025

Question 6

Create full date from parts using DATE_FROM_PARTS()

Syntax

```
Select member_id,
    DATE_FROM_PARTS(start_year, start_month, start_day) AS full_start_date
  from memberships;
```

Member_id	full_start_date
1	2023-05-10
2	2022-11-25

Question 7

Extend each renewal-date by 1 year using DATEADD().

```
Select sub_id,
        plan,
DATEADD(year, 1, renewal_date) AS extended_renewal_date
  from subscriptions;
```

sub_id	Plan	extended_renewal_date
11	Basic	2026-01-01
12	Premium	2026-03-15

Question 8

Show current date and difference from order_date. Use current_date and Datediff().

Syntax

```
Select order_id,
        order_date,
  current_date AS today_date,
    Datediff (day, order_date, current_date) AS days_since_order
    from Orders
```

order_id	order_date	today_date	days_since_order
1001	2025-04-15	2026-04-22	372
1002	2025-04-10	2026-04-22	377

Question 9

Extract the year from training_date using DATE_Part() or Extract()

Syntax

```
Select training_id,
        topic,
      Year (training_date) AS training_Year
  from trainings;
```

training_id	topic	training_Year
1	Safety	2025
2	Compliance	2025

Question 10

Extract hour and minutes from published_on

Syntax

```
Select Post_id,
        title,
    DATEPART (hour, Published_on) AS hour_Published,
    DATEPart (minute, Published_on) AS minute_published
    From Blog_Posts;
```

Post_id	title	hour_Published	Minute_Published
1	SQL tips	10	15
2	Data Cleaning	16	45

Question 11

Calculate days left until license expiry using DATEDIFF() and todays date

Syntax

```
Select driver_id,
        license_expiry,
    DATEDIFF (day, current_date, license_expiry) AS days_left
  from Drivers;
```

driver_id	license_expiry	days_left
301	2025-08-10	-256
302	2023-12-31	-843

Question 12

Display the current timestamp and calculate seconds since the message was sent

Syntax

```
Select  message_id,
        sent_timestamp,
    current_timestamp AS current_timestamp,
    DATEDiff (second, sent_timestamp, current_timestamp) AS seconds_since_sent
  from Messages;
```

Message_id	Sent_timestamp	current_timestamp	Seconds_since_sent
1	2025-04-19 09:32:45	2025-04-20 10:00:00	88275
2	2025-04 18 23:59:59	2025-04-20 10:00:00	122401

Question 13

Add 15 days to return_date using DATEADD() to show restock_date

Syntax

```
Select  return_id,
        return_date,
    DATEADD (day, 15, return_date) AS restock_date
  from Returns;
```

return ID	return-date	restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Question 14

Convert assigned_on to date using TO_DATE() (if is stored as string)

Syntax

~~Select assign_id,~~

~~CAST(ASsig~~

```
Select assign_id,
    TO_DATE(Assigned_on, 'YYYY-MM-dd') AS assigned_on_date
from assignments;
```

assign_id	assigned_on_date
1	2025-03-01
2	2025-03-05

Question 15

Convert Scheduled_time to formatted string like 'April 19, 2025 at 2:00 PM' using To_Char()

```
Select meeting_id,
    TO_Char(Scheduled_time, 'FMMonth DD, YYYY "at" FMHH:MI AM') AS
    Formatted_meeting_time
From Meetings;
```

Meeting_id	Formatted_meeting_time
1	April 19, 2025 at 2:00 PM
2	April 19, 2025 at 9:30 am